

Total No. of Questions : 8]

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Roll No

EC-7004(3)-CBGS

B.E. VII Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Digital Image Processing

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions out of eight questions.
ii) All questions carry equal marks.
iii) Assume suitable data, if required.

1. a) Give a brief introduction on Digital Image Processing. Give some examples of fields that use DIP.
b) Discuss some of the parameters of Image processing system. What do you mean by Gray Level and resolutions?
2. a) Explain the different elements of visual perception.
b) What is Image transform? What are the applications of transform?
3. a) Write the expression of 2D continuous Fourier transform pair.
b) Write the expression for Hadamard transforms. Also write the properties of Hadamard transform?
4. a) Define of KL transform. Justify that KLT is an optimal transform.
b) Name the categories of Image Enhancement and explain.

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5.
 - a) Explain the fundamentals of spatial filtering with suitable examples.
 - b) Define Histogram. What is a Median filter and derivative filters?
6.
 - a) Give a brief introduction on Image Restoration. Explain its noise models.
 - b) What do you mean by linear position invariant degradations? Explain it with suitable examples.
7.
 - a) Discuss some of the fundamentals of Image compression. Discuss some of the basic compression methods.
 - b) What is the concept behind algebraic approach to restoration? Explain with the help of suitable example?
8. Write short notes (any four):
 - a) Digital Image Watermarking
 - b) Golomb coding
 - c) Walsh Transform
 - d) Image sampling and Quantization
 - e) Wiener filtering
